

YERMAKOVSKOYE, Russian Federation

WMO: 298690

Lat: 53.270N

Lon: 92.403E

Elev: 298

StdP: 97.80

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.7	-33.4	-38.7	0.1	-36.6	-35.5	0.1	-33.3	8.6	-7.3	7.1	-8.3	0.0	50	0.549

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	30.0	19.8	28.3	19.1	26.5	18.2	21.2	27.8	20.2	26.4	19.3	24.7	1.6	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	14.2	24.0	18.0	13.4	23.2	17.2	12.7	22.2	63.1	28.0	59.6	26.4	56.5	24.7	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.0	5.7	4.6	-38.7	33.0	3.6	1.3	-41.3	33.9	-43.5	34.7	-45.5	35.4	-48.2	36.3		
			-38.2	23.4	3.7	1.3	-40.9	24.4	-43.1	25.1	-45.1	25.9	-47.8	26.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.4	-17.8	-13.8	-3.8	4.8	11.1	17.5	19.4	16.9	10.3	2.9	-5.9	-14.1
	DBStd	13.95	8.47	8.09	6.55	4.84	4.57	3.72	2.79	3.42	4.06	4.61	6.98	8.20
	HDD10.0	3661	862	668	428	165	43	1	0	1	44	223	478	748
	HDD18.3	5941	1120	901	686	406	228	57	21	68	242	478	728	1006
	CDD10.0	878	0	0	0	9	77	228	291	216	54	3	0	0
	CDD18.3	116	0	0	0	0	3	34	53	24	1	0	0	0
	CDH23.3	1318	0	0	0	7	99	416	481	283	32	1	0	0
	CDH26.7	338	0	0	0	1	21	115	122	72	7	0	0	0
Wind	WSAvg	1.3	0.8	1.0	1.6	1.7	1.7	1.4	1.1	1.0	1.2	1.2	1.3	1.1
Precipitation	PrecAvg	544	18	15	17	35	49	77	82	80	65	45	35	25
	PrecMax	804	39	34	49	67	100	135	174	183	233	129	89	61
	PrecMin	397	4	3	5	4	4	28	43	27	15	9	9	12
	PrecStd	94	9	8	10	16	23	28	28	35	41	23	19	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.4	4.0	13.7	23.7	29.2	32.0	31.5	31.3	27.1	19.3	10.0	2.5
		MCWB	-0.9	0.4	6.0	11.7	15.8	19.4	21.6	21.6	17.9	11.3	5.2	-0.3
	2%	DB	-1.0	1.7	10.4	19.9	26.3	29.6	29.7	28.7	23.0	15.1	7.0	0.4
		MCWB	-2.9	-1.1	4.6	10.1	14.1	18.9	20.6	20.2	15.5	9.4	3.7	-2.0
	5%	DB	-3.1	-0.2	7.7	16.9	23.7	27.7	28.0	26.5	20.3	12.1	4.7	-1.0
		MCWB	-4.7	-2.3	3.0	8.9	13.1	18.3	19.9	19.0	14.2	7.8	2.1	-2.9
	10%	DB	-5.6	-2.4	5.2	13.9	21.0	25.7	26.1	24.4	17.8	10.0	2.8	-3.1
		MCWB	-6.9	-4.4	1.4	7.4	12.2	17.5	19.1	17.9	12.9	6.7	0.7	-4.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.1	1.1	7.0	12.5	17.4	21.5	22.9	22.3	18.5	12.1	6.0	0.1
		MCDB	0.9	3.2	12.7	21.8	26.9	28.9	29.6	29.6	24.4	17.5	8.9	1.6
	2%	WB	-2.7	-0.8	4.9	10.8	15.6	20.1	21.5	20.7	16.4	10.0	4.0	-1.6
		MCDB	-1.1	1.5	9.9	18.1	23.7	27.5	27.9	27.3	21.9	14.0	6.6	0.1
	5%	WB	-4.7	-2.5	3.2	9.3	14.1	19.1	20.6	19.5	14.9	8.3	2.4	-2.9
		MCDB	-3.2	-0.2	7.4	15.7	21.5	25.6	26.4	25.4	19.3	11.8	4.6	-1.2
	10%	WB	-6.9	-4.3	1.7	7.7	12.7	18.1	19.7	18.4	13.4	6.8	0.7	-4.4
		MCDB	-5.5	-2.4	5.0	13.4	19.2	24.0	24.9	23.3	16.9	9.6	2.6	-3.1
Mean Daily Temperature Range	5% DB	MDBR	10.5	12.2	12.3	12.0	12.6	12.1	10.7	11.2	11.4	9.7	7.8	8.9
		MCDBR	8.8	11.8	15.1	18.9	18.9	16.4	14.2	15.5	16.9	15.0	9.4	8.5
		MCWBR	7.3	9.2	9.6	9.9	8.2	6.9	6.2	7.4	9.3	9.0	6.4	6.8
	5% WB	MCDBR	8.4	11.4	14.0	17.0	15.3	13.6	12.4	13.9	15.3	14.2	8.7	8.1
MCWBR		7.1	8.9	9.1	9.4	7.2	6.0	5.7	7.0	8.8	8.9	6.1	6.7	
Clear-Sky Solar Irradiance	taub	0.284	0.320	0.367	0.411	0.418	0.403	0.413	0.399	0.363	0.350	0.325	0.290	
	taud	2.056	2.012	1.985	2.122	2.178	2.329	2.326	2.338	2.375	2.277	2.102	2.005	
	Ebn at Noon	654	747	792	813	830	849	832	821	808	722	592	549	
	Edh at Noon	74	108	138	138	137	119	118	110	94	83	72	65	
All-Sky Solar Radiation	RadAvg	1.07	2.11	3.50	4.43	5.20	6.05	5.51	4.59	3.25	1.84	0.94	0.71	
	RadStd	0.09	0.14	0.23	0.27	0.46	0.60	0.41	0.37	0.33	0.17	0.12	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page